Who Discovers the Price?

A real cross-venue price-discovery lead — and why a market-maker can't trade it. Tick-level evidence from the 2026 World Cup.

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The headline of this note is a non-result, and that is the point. Using tick-level order-book data from 86 World Cup matches on Kalshi and Polymarket, I find a genuine cross-venue price-discovery lead — on a goal, Polymarket's mid moves ***~600 ms*** *first and carries a* ***~81%*** *Gonzalo–Granger information share, leading* ***61 of 63*** *cointegrated matches (computed on mid-quotes, which sidesteps the ~59% trade-direction problem documented for Polymarket). Conditioned on the news the lead is sharper still:* ***~86%*** *of price discovery in goal windows versus ~53% in calm play.* ***And then it is not tradeable.*** *At the goal the book vanishes — spread* ***~8***× *wider, best-price depth under* ***1%*** *of normal — so the median 12.0-cent stale-quote gap a reactor would lift carries a median +10.8c net on paper but is* ***not harvestable at size****: gated on the depth actually resting in the book, the* ***median match yields no harvestable goal at all*** *(harvestable goals exist — ~11% goal-weighted on the 21-match subset still reconstructible from the archive — but they are a minority, clustered in a minority of matches, and unidentifiable in advance). That collapse-and-refill is the market-maker pulling quotes against toxic, information-motivated flow:* ***adverse selection observed in real time****, and the reason the lead is a property of price discovery, not a free trade. Forecasts for the wider project were pre-registered before kickoff and graded publicly on 19 July — verifiable without trusting the author.*

Summary.

- **Claim:** Polymarket is the price-discovery leader of the Kalshi–Polymarket pair on the same World Cup outcomes.
- **Independent methods agree:** an event-study reaction lead, Hasbrouck / Gonzalo–Granger information shares, an order-flow mechanism check, and a cross-sectional level check against the sharp line.
- **Numbers:** lead-lag **281 vs 111** events to Polymarket (**72%**, median **+600ms**) across 79 matches, and **57 of 66** matches lean Polymarket (sign-test $p \approx 1.2 \times 10^{-9}$, cluster-robust); Gonzalo–Granger share **~81%**, leading **61/63** cointegrated matches; OFI is a within-venue mechanism (no cross-venue flow lead); mean gap to the sharp line **0.15pp** (Poly) vs **0.26pp** (Kalshi) at 2026-06-22.
- **Anatomy of a goal:** discovery concentrates at the news (Polymarket **~86%** in goal windows vs **~53%** calm); the book withdraws hardest where it leads (spread **~8×**, depth **<1%**); so the median 12.0c stale-quote gap is a median +10.8c on paper, yet the **median match** yields no harvestable goal once gated on the vanished depth.
- **Honest caveats:** not tradeable after cost (sub-second + the book withdraws); one tournament, one sport; event-clustering means the formal significance unit is the match, not the event.
- **The maker's view:** the collapse-and-refill is the quoter's defence against toxic flow — a maker who held stale quotes through the goal would be the one adversely selected. The unharvestable gap is exactly the loss the withdrawal avoids.
- **Verifiable, not trust-me:** the project's forecasts were pre-registered before kickoff and were graded publicly on 19 July — 6 pass, 2 fail, 3 inconclusive, hits and misses — a track record that can't be edited after the fact.
- **Why it matters:** it answers a question a 2026 Polymarket microstructure study explicitly left open (on live in-play sports, not the election markets prior work used), and it tells anyone aggregating prediction-market signals which venue to trust first — while showing, with the costs in, that the lead is information not alpha.

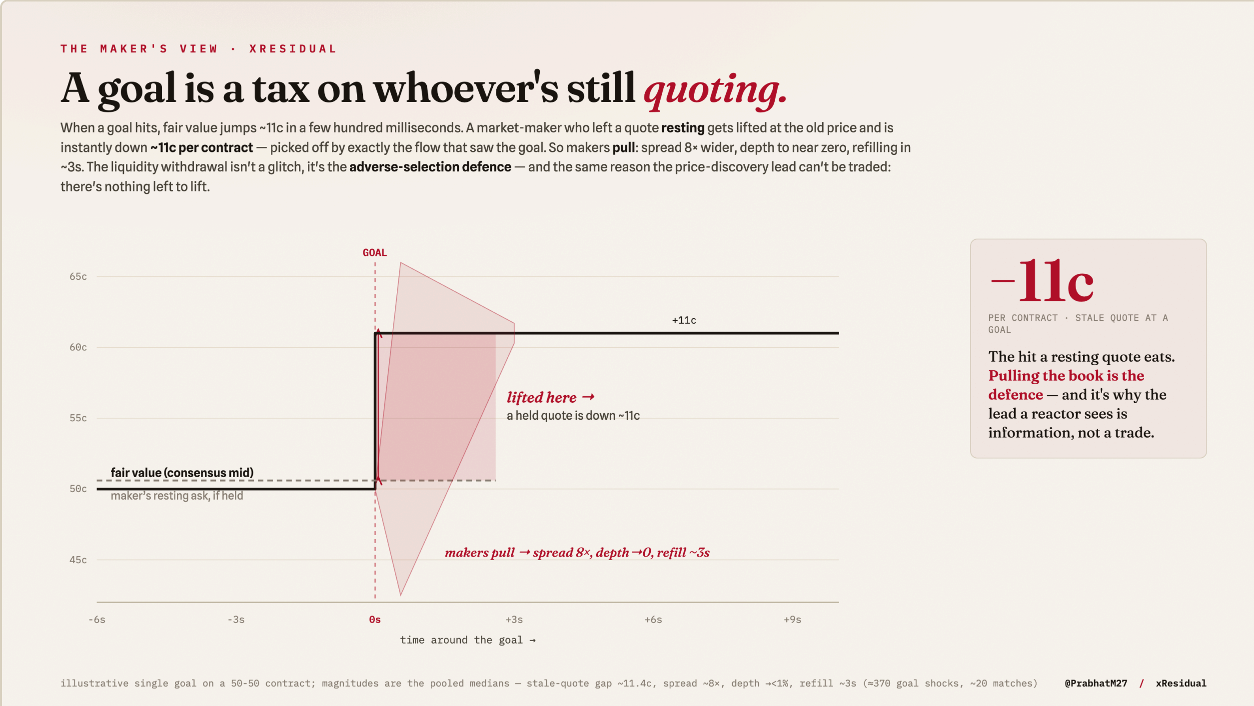

Figure 1. The anatomy of a goal, from the maker's side. Fair value jumps ~11c in a few hundred milliseconds. A quote left resting in the book is lifted at the stale price and is down ~11c per contract — so makers pull: spread ~8× wider, depth to near zero, refilling in ~3s. That withdrawal is the adverse-selection defence, and the reason the price-discovery lead a reactor sees is information, not a tradeable edge. (Illustrative single goal on a 50–50 contract; magnitudes are the pooled medians across ~370 goal shocks.)

1. The open question

Kalshi (US-regulated) and Polymarket (crypto-native) quote the *same* World Cup outcomes through very different microstructure — different fee and tick regimes, user bases, and settlement. That makes the pair a natural laboratory for cross-venue price discovery: when new information arrives, which venue moves first, and which merely follows?

A 2026 study of the Polymarket order book [1] maps that venue's microstructure in detail but states plainly that a cross-venue comparison against Kalshi — the price-discovery question — is left for future work. This note answers it for a clean, repeated, exogenous information event: goals in the 2026 World Cup. I form no strong prior on direction and let the data decide.

2. Data

I capture both venues' full level-2 order books and every trade over WebSocket, at tick resolution, from an always-on collector running 24/7 through the tournament. For each match I pair contracts that reference the identical outcome (one Kalshi ticker and one Polymarket token per team and the draw), de-vig each venue's quotes to probabilities, and align the two mid-price series on a common clock. The event study runs on **79 matches**, OFI on **83 matches**, and the information-share estimates (below) on the **63 cointegrated matches** (104 contracts) whose cross-venue spread passes the ADF guard (final sample, through the 19 July final).

Coverage. In-play capture is near-complete: **86 of 102 completed matches** on both venues (the quarter- and semi-finals included), of which the cointegrated subset feeds the information-share estimates. **Four** of the earliest knockout ties — Germany–Paraguay, Ivory Coast–Norway, France–Sweden, USA–Bosnia (29 June – 2 July) — have no cross-venue capture, and a handful of group games were captured on one venue only; the background snapshot and bookmaker-odds series ran throughout. The gap changes which matches populate the in-play samples, not the direction of the result.

3. Methods

I measure the same phenomenon three ways, at increasing rigor, so the conclusion does not rest on any single technique.

3a. Event-study reaction lead

I detect repricing shocks (goals) on each venue's mid series and cross-correlate the two around each event to estimate which venue's mid moves first, to the millisecond. Intuitive, but timing in a sub-second window is noisy.

3b. Information shares (the core)

The two de-vigged mids for one outcome are cointegrated with the natural vector (1,−1): arbitrage keeps their difference stationary. I test that with an Augmented Dickey–Fuller test and *discard any contract that fails* (on a non-cointegrated pair the shares are a spurious-regression artifact). On the survivors I fit a bivariate vector error-correction model and report the **Hasbrouck** [3] information-share bounds and the ordering-free **Gonzalo–Granger** [4] permanent-component share. The intuition: the venue that adjusts *less* to the spread (the smaller error-correction coefficient) is the one others converge toward — the price-discovery leader.

3c. Mechanism

I reconstruct order-flow imbalance (OFI) from top-of-book changes [5] and test (i) whether OFI moves each venue's own price — the canonical within-venue result — and (ii) whether one venue's flow predicts the *other's* next move. I also recompute the lead on the imbalance-weighted microprice [6] rather than the mid.

4. Results

The winner market is efficient. As a control, the de-vigged outright-winner prices agree across venues (and with a calibrated model) to within **~0.15 pp**. There is no cross-venue mispricing in the liquid market; the effect below is about *speed of discovery*, not a level disagreement.

MEASURE	SAMPLE	RESULT
Reaction lead (event study)	79 matches, 392 events	Polymarket first in **281**, Kalshi in **111** (**72%**), median **+600ms** among Polymarket-led events (signed pooled median +400ms, IQR [–200, +800])
Reaction lead, per match (cluster-robust)	66 matches	**57 of 66** lean Polymarket (sign-test $p \approx 1.2\times10^{-9}$; design effect 1.13)
Gonzalo–Granger share	63 cointeg. matches	Polymarket **~81%**, leads in **61 / 63** ($p \approx 4.4\times10^{-16}$)
Hasbrouck share (band)	104 contracts	Polymarket band **~77–92%**
OFI → own-venue price impact	83 matches	strong within-venue, both directions (bin-level t; see §6)
OFI cross-venue lead	83 matches	**null —** no clean asymmetry either direction

All three methods point the same way: Polymarket leads. The mechanism check sharpens the interpretation — order flow clearly drives price *within* each venue (textbook [5]), but there is *no* cross-venue flow lead. The lead therefore lives in the *price*, not in observable flow: Polymarket's mid simply updates to new information first.

5. The anatomy of a goal

The results above are pooled over each match. Conditioning on the actual information event — a goal — turns the single ~81% into a dynamic picture, and answers the question any price leader should be asked: is the lead a tradable edge?

MEASURE (PER GOAL)	SAMPLE	RESULT
Information share, event-conditional	~20 matches	Polymarket GG **~86%** in goal windows vs **~53%** in calm play (Hasbrouck ~82% vs ~56%)
Liquidity at the goal (top of book)	~370 shocks	spread **~8×** (Poly) / **~2×** (Kalshi); best-price depth falls to **<1%** of normal; refill ~3–4s
Harvestability of the lead	405 goals / 66 matches	median gross stale gap 12.0c, median net **+10.8c on paper** after spread — but the **median match yields no harvestable goal** once gated on depth (net-positive on paper, no depth to lift; goal-weighted 11.1% on the 21-match reconstructible subset)
Reaction size vs fair value	8 clean matches, 22 gated goals	market books **~0.35×** the model's fair log-odds jump; under-shoots in **22 of 22** goals and **8 of 8** matches (sign p 0.008) — an **under-reaction** (preliminary)

Discovery concentrates at the news. Away from goals the two venues are near-coequal (~53/47); the instant a goal lands, Polymarket carries ~86% of the permanent price move. The flagship lead is not a steady background hum but a *news-event* phenomenon — Polymarket leads precisely because it incorporates the goal first.

And the same venue withdraws liquidity the hardest. At the goal, makers pull their quotes: the spread blows out (~8× on Polymarket, ~2× on Kalshi) and best-price depth collapses to under 1% of normal, refilling only after a few seconds. Unlike a betting exchange — Betfair suspends and cancels resting orders on a goal (Croxson & Reade 2014) — Kalshi and Polymarket never halt, so this is *informational* liquidity withdrawal, not a mechanical suspension.

So the lead is real but not systematically tradeable. A naive ledger makes the stale follower look like free money: a median 12.0c of staleness and a median +10.8c net of the spread, on every goal (medians across the 66 ledger matches, not a mean and not a per-goal average). But that quote has vanished — depth at the goal is ~0.5% of normal — so once gated on what is actually resting in the book, the **median match yields no harvestable goal**. That figure is a median across the 66 ledger matches, and the precise reading matters: the typical match offers a follower nothing, not that no goal anywhere was capturable. On the portion of the per-game ledger still reconstructible from the archive (**21 matches, 117 goals**) the goal-weighted rate is **11.1%**, concentrated in **6 of those 21** matches. So the claim is distributional, not absolute: a minority of goals, in a minority of matches, that a follower cannot pick out in advance because the depth is gone at the instant the signal fires. "The news is in the price before you can trade it" (Croxson & Reade 2014) holds here for a precise, measured reason: not slow pricing, but a book that empties in the same instant it reprices.

And it under-reacts to the goal it just priced. Conditioning the same event on a fair-value benchmark — a calibrated clock-and-Poisson model anchored to the pre-match price — the market's post-goal move is a median **~0.35**× the model's fair jump *in log-odds* (so this is not the logistic curve flattering the result; the raw-probability "value of a goal" curve and an apparent "underdog goals move twice as far" both dissolve once that geometry is removed, and I do not claim them). The market under-shoots the fair move in every one of the **22** goals whose quote moved and in **8 of 8** cleanly-reconstructed matches, and the move sticks (60-second reversion ~0), so it is a persistent under-reaction rather than an overshoot. A decisive check rules out the alternative — that my benchmark merely over-reacts: the model's larger post-goal probability is the *better* forecast of the eventual result (log-loss 0.235 vs 0.361; Brier 0.073 vs 0.113), so the larger jump was warranted. I report this as **suggestive, not firm** — the cluster-honest unit is the match, and 8 of 8 is a sign-test $p = 0.008$ on eight matches; on the wider sample that includes shock-inferred goal times the direction holds (**54 matches, 185 goals**; mean under-shoot **3.1pp** in P(home win)) but the outcome test muddies, which is why I trust only the clean-reconstruction subset. The knockout captures did not resolve it: the clean-reconstruction subset stayed at 8 matches, so this remains the weakest-supported result here.

6. Robustness

Immune to the trade-direction problem. [1] shows that trade direction inferred from Polymarket's public feed agrees with on-chain ground truth only ~59% of the time, which corrupts any signed-trade measure (effective spread, Kyle's λ, trade-sign OFI). Every estimate here is computed on *mid-quote* moves and book changes, not signed trades, so it is unaffected; I verified our signed-flow code is used by none of these results.

No spurious shares. The ADF gate drops non-cointegrated pairs, so a Hasbrouck/GG number is only reported where the (1,–1) spread is genuinely stationary.

Significance, stated honestly. Goal events cluster within matches and are not independent, so the naive event-level binomial ($p \approx 4 \times 10^{-18}$ on 281 of 392) and the bin-level OFI t-statistics overstate significance. The unit that carries the claim is the *match*: 61 of 63 cointegrated matches show a Polymarket lead, with a ~81% central share. That match-clustered bootstrap is now run (`scripts/harden_leadlag_stats.py`): the clustering is mild (ICC 0.032, design effect 1.13), the reaction-lead 95% CI holds at [67%, 76%], and the per-match sign-tests are decisive — **57 of 66** matches lean Polymarket on the reaction lead ($p \approx 1.2 \times 10^{-9}$) and **61 of 63** on the information share ($p \approx 4.4 \times 10^{-16}$). The qualitative result is not in doubt.

7. What this is *not*

Not a trade. The lead is a few hundred milliseconds. Documented cross-venue arbitrage exists (~$40M, 2024–25) but is seconds-lived and fee-bounded; a 600 ms head start vanishes inside the bid–ask once costs are crossed. This is price-discovery *leadership*, a property of market structure — not free money.

Scope. One tournament, one sport, a ~three-week window, 63 cointegrated matches. Polymarket's documented wash trading (~25% of historical volume) is a caveat on volume-based measures, less so on mids. The Hasbrouck band is wide where the two venues' innovations are correlated, which is why I lead with the ordering-free Gonzalo–Granger number.

8. Related work & contribution

The within-venue OFI→price relation reproduces Cont–Kukanov–Stoikov [5]; the microprice's edge over the mid follows Stoikov [6]; the information-share machinery is Hasbrouck [3] and Gonzalo–Granger [4]; an independent 2025 study finds a Polymarket discovery lead in other markets [2], which this corroborates — but where [2] reads that lead as "economically meaningful arbitrage," the depth-gated ledger here qualifies it: on a clean repeated-event stream the lead is real, yet the book withdraws in the same instant it reprices, so the stale-quote gap yields nothing to a follower in the median match and the arbitrage is *not* economically meaningful net of the cost of immediacy. The contribution is, to my knowledge, the first **cross-venue Kalshi–Polymarket information-share estimate at tick resolution on a clean, repeated event stream** (goals), robust to the trade-direction problem [1], paired with a depth-gated harvestability test that separates a real price-discovery lead from a *tradable* one.

9. Conclusion

On identical World Cup outcomes, Polymarket discovers price first: it reacts ~600 ms sooner and carries ~81% of the permanent price movement, in 61 of 63 cointegrated matches measured. The practical implication is narrow but real — an aggregator of prediction-market signals should weight Polymarket's mid as the leading indicator and Kalshi's as the lagging confirm — and the trading implication is honestly none, after costs. The World Cup was chosen as the testbed for a practical reason — a dense, scheduled stream of discrete information events across two liquid venues is the cleanest setting to measure discovery — but nothing in the method is sport-specific: the same information-share and tick-capture stack applies to the macro, rate, and event contracts that are becoming these venues' durable book, which is where I would point it next. Everything is reproducible from the public repository; forecasts in the wider project are committed to a forward-only ledger, and the core estimation math is frozen, so none of this is fit after the fact.

References.

1. "The Anatomy of a Decentralized Prediction Market: Microstructure Evidence from the Polymarket Order Book," 2026, arXiv:2604.24366.
2. Ng, Peng, Tao, Zhou, "Price Discovery and Trading in Modern Prediction Markets," 2025 (SSRN 5331995).
3. Hasbrouck, J. (1995), "One Security, Many Markets: Determining the Contributions to Price Discovery," *Journal of Finance*.
4. Gonzalo, J. & Granger, C. (1995), "Estimation of Common Long-Memory Components in Cointegrated Systems," *JBES*.
5. Cont, R., Kukanov, A. & Stoikov, S. (2014), "The Price Impact of Order Book Events," *Journal of Financial Econometrics*.
6. Stoikov, S. (2017), "The Micro-Price: A High-Frequency Estimator of Future Prices," SSRN 2970694.

xResidual — a live study of how prediction markets price the 2026 World Cup. Methods, data pipeline, and the live findings dashboard: github.com/Thesavagecoder7784/xResidual